A248 - Magneto-optic trap

Ben Nam-Soo Weihrauch[1]
[1] *University of Bonn*

, Christian Gottschlich[2]
[2] *University of Bonn*

(Dated: November 10, 2026)

We set up a magneto-optical trap for ^{85}Rb atoms in a vapour cell, using two grating-stabilized diode lasers locked to the $F = 3 \to F' = 4$ cooling and the $F = 2 \to F' = 3$ repumping transitions via polarization spectroscopy. After aligning the two horizontal MOT axes and locking the lasers, we characterized the trap with a powermeter, a CCD camera and a photodiode. From a 2D Gaussian fit to fluorescence images we obtained a cloud size of $\Delta x = 2.4(12)\,\mathrm{mm}$ and $\Delta y = 3.5(18)\,\mathrm{mm}$, where the dominant uncertainty comes from an estimated camera-object distance correction. The dependence of the fluorescence on the $\lambda/4$ plate angle followed an adapted Malus's law with an optimum at $\theta_{\mathrm{opt}} = 102(2)^\circ$, and rotating the retro-reflected waveplates had no measurable effect. Varying the anti-Helmholtz coil current yielded a linear fluorescence response above 2 A with an extrapolated turn-on current of $I_c = 1.85(5)\,\mathrm{A}$. From three loading curves we determined a characteristic loading time of $\tau = 1.79(10)\,\mathrm{s}$, from which we derived a Rb-Rb collisional cross section of $\sigma_{\mathrm{Rb}} = 1.60(9) \times 10^{-18}\,\mathrm{m}^2$ at a chamber pressure of $p \approx 5.27 \times 10^{-6}\,\mathrm{Pa}$. Scanning the cooling laser frequency, the MOT fluorescence peaked at a detuning of $\delta_{\mathrm{opt}} = -8.5(1)\,\mathrm{MHz}$, corresponding to approximately $1.4\,\Gamma$ red of the $F = 3 \to F' = 4$ transition. From the measured fluorescence power, the beam profile and the collection geometry we estimated an atom number of $N = (1.3 \pm 0.3) \times 10^6$, with the dominant uncertainty coming from the distance between the collection lens and the trap.

I. INTRODUCTION

The Magneto-Optical Trap (MOT) is a fundamental experimental tool used to cool and confine neutral atoms to temperatures in the microkelvin regime [1] . The technique relies on the continuous scattering of photons from near-resonant laser beams, which transfer momentum to the atoms. To achieve cooling, the laser frequency is red-detuned from the atomic resonance. Due to the Doppler effect, an atom moving toward a laser beam perceives the light as blue-shifted toward resonance, leading to an increased photon absorption rate. When a photon is absorbed, the resulting momentum transfer gives the atom a "kick" in the opposite direction, effectively decreasing its velocity.

Because the atoms subsequently decay via isotropic spontaneous emission, the net achievable cooling is finite, and cooling to absolute 0 K is not possible. This limit is known as the Doppler-limit [1] . By using three mutually orthogonal, counter-propagating laser beam pairs, this cooling effect acts in all three spatial dimensions. This method is called optical molasses [2].

While optical molasses cools the atoms, it provides no spatial confinement. To trap the atoms, the MOT introduces a spatially dependent restoring force using a spherical magnetic quadrupole field generated by a pair of coils in an anti-Helmholtz configuration. This magnetic field gradient induces a position-dependent Zeeman-shift in the atomic energy levels. By utilizing appropriate circular polarization for the opposing laser beams according to quantum selection rules, atoms displaced from the zero-field center preferentially absorb photons from the laser beam pointing back toward the origin. The exact theory combining this spatial and velocity dependence is further detailed in [2, 3] .

II. EXPERIMENTAL SETUP

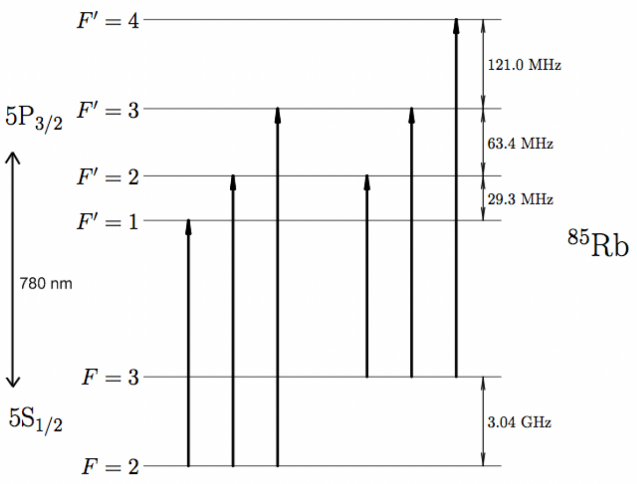


FIG. 1: Relevant energy level scheme for the ^{85}Rb isotope used in the experiment. The solid arrows indicate the primary cooling transition ($F = 3 \to F' = 4$) and the necessary repumping transition ($F = 2 \to F = 3$). Image taken from [4].

The experiment captures and cools the ^{85}Rb isotope from a thermal background vapor. We utilize the $5^2S_{1/2}, F = 3 \rightarrow 5^2P_{3/2}, F' = 4$ hyperfine transition for the primary cooling cycle, as illustrated in Figure 1 . As these transitions correspond to the infrared regime of wavelengths, both lasers used have wavelengths in the infrared. Because quantum selection rules for the $F' = 4$ decay dictate $\Delta F = 0, \pm 1$, the excited $F' = 4$ state can only decay back to the $F = 3$ ground state, establishing an approximately closed two-level cooling system. However, off-resonant excitation to the $F' = 3$ state occurs at a finite rate [4] . Atoms in the $F' = 3$ state can then spontaneously decay into the $F = 2$ ground state, which is transparent to the cooling laser. To prevent the continuous loss of atoms into this "dark state", we employ a secondary repumping laser tuned to the $F = 2 \rightarrow F' = 3$ transition, continuously returning lost atoms to the cooling cycle.

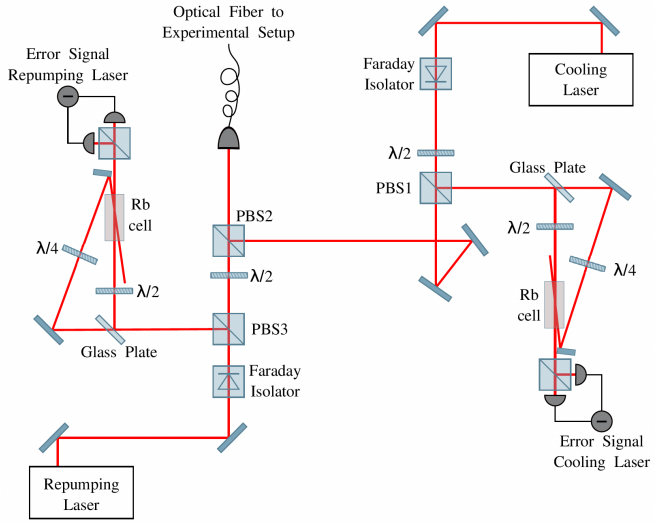


FIG. 2: Optical setup for generating, stabilizing, and combining the cooling and repumping laser frequencies. One error signal for each Laser is created using polarization spectroscopy [4] , which is used for locking the Laser beams to a given frequency. The cooling and repumping laser are combined and send through an optical fiber to the experimental vacuum chamber setup. Image taken from [4].

Both the cooling and repumping laser beams are generated by independent grating-stabilized diode lasers. To ensure the frequencies remain stable at the required detunings, we derive error signals using polarization spectroscopy [4] (a Doppler-free technique) and actively lock the lasers via an electronic feedback loop. As shown in Figure 2, the stabilized cooling and repumping beams are spatially overlapped using a polarizing beam splitter (PBS) and subsequently coupled into a single-mode optical fiber, which transports the light to the vacuum chamber.

Upon exiting the fiber the beam is split into three

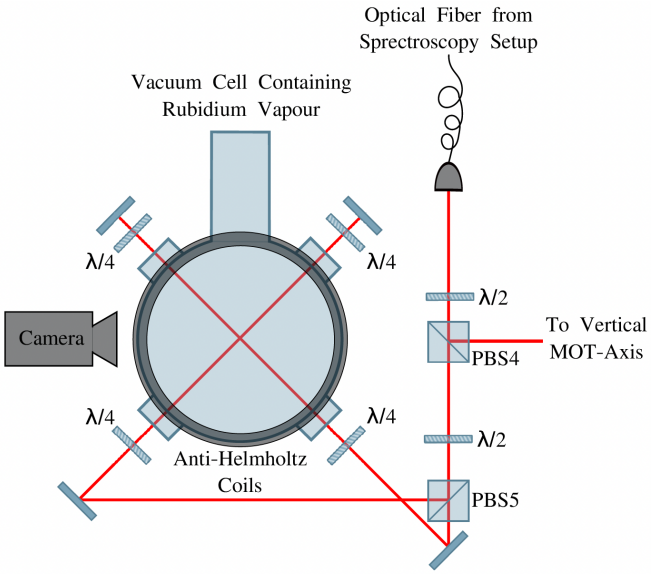


FIG. 3: Top-down view of the vacuum chamber and the retro-reflection optics creating the six-beam MOT configuration. Image taken from [4].

paths, the 2 in the horizontal plane have the same intensity while the vertical beam has a higher intensity since it also has to counteract gravity. As depicted in Figure 3 , these three beams are directed along mutually orthogonal axes through the center of the vacuum chamber, where the anti-Helmholtz coils generate the required magnetic field gradient. Before entering the chamber, each beam passes through a $\lambda/4$ waveplate to establish the necessary circular polarization. After passing through the chamber, each beam traverses a second $\lambda/4$ plate, reflects off a mirror, and passes through the waveplate a third time. This double-pass retro-reflection automatically reverses the propagation direction and flips the polarization handedness, providing the correct counter-propagating beam dynamics for the MOT without requiring six independent laser sources. Additionally, we monitor the trap using a CCD camera for spatial size characterization and a calibrated photodiode to continuously measure the emitted atomic fluorescence. Finally, we use an infrared (IR) camera as well as an IR card to calibrate the mirrors of the vacuum chamber to ensure proper spatial alignment of all six orthogonal beams.

III. MOT ADJUSTMENT

This section describes how we aligned the MOT beams, locked the lasers, and found and optimised the trap.

Pre-set elements that we did not change

Two parts of the setup that, in principle, belong to the alignment were already correctly set when we started, so we did not touch them:

- **Power balancing of the MOT axes.** In general, the three MOT axes should carry equal cooling-beam power, which is done by rotating the $\lambda/2$ plates in front of PBS4 and PBS5, shown in Figure 3 , with the repumping laser switched off. In our case, this had been done before we began, and we did not readjust it.
- $\lambda/4$ **plates in front of the chamber.** These set the circular polarisation (σ^+/σ^-) of the MOT beams, which is essential for the trap to work. We left these plates at the orientation we found them in. (Their effect on the MOT was studied separately in Section V.)

Beam alignment

We only aligned the two horizontal MOT axes (which we will call x and y). The vertical z-axis beam was left untouched, as its alignment was already correct from the previous use of the setup.

The roughly 780 nm light of the combined laser beam [4] is invisible to the eye, and the full beam is several millimetres wide. Therefore, we could not just look at the beam and judge whether it was centered or not. To aid, we used three aids:

- **An iris** placed in the beam was closed down to a small diameter. This made the beam more thin, making the alignment easier. After the alignment was finished, we opened the iris up to its full diameter such that the full beam went through the chamber, which is what we need for a large enough capture volume.
- **A target with a small hole stuck to the entrance window** of the vacuum chamber. This gave us a fixed reference point right at the chamber: when the narrow beam from the iris passed through this hole, we knew the beam was hitting the chamber at the correct spot.
- **A height-adjustable aperture on a magnetic stand.** We taped an IR-sensitive card with a small hole to a folding ruler, and taped the rule to a magnetic base sitting on the optical table. This gave us a hole at any height we wanted that stayed in place by itself, so we did not have to hold a card by hand while checking the back-reflection. We preferred this over holding the card by hand, because small hand movements made it hard to tell whether the back-reflection was actually overlapping the forward beam.

For each of the two horizontal axes, we did the following:

1. Close the iris down to a narrow beam. Steer the forward beam with the two mirrors so that it passes through the small hole in the target on the entrance window and through the centre of the opposite chamber window. The IR camera and the IR viewer were used to see where the beam was hitting.

2. Adjust the retroreflecting mirror on the far side of the chamber so that the back-reflection runs back along the forward beam all the way to the last mirror before the vacuum chamber. We checked this with the mounted hole card at several positions along the path: if the back-reflection went through the small hole at every position, the two beams were overlapping.

Additionally, the two horizontal axes had to cross at the same point in the middle of the chamber. We did not get this right in one go: we went back and forth between the x and y axes several times, each time bringing the two crossing points a bit closer together, until on the infra red camera the beams looked like they were overlapping in the vacuum chamber.

Finding the MOT

With the beams roughly aligned, the anti-Helmholtz coils were switched on at a nominal value of $I = 2.5\,\mathrm{A}$ and we opened the iris back up to the full beam diameter. From this point on, all further beam alignment was done with the full beam, since the optimisation signal was now the brightness of the MOT itself.

Both lasers were initially in scan mode with the scan amplitude on its maximum. The diode current and the scan offset were varied until the rubidium absorption spectrum appeared on the oscilloscope. The peaks were identified by comparison with the reference spectra at the end of the script [4] : the $^{85}\mathrm{Rb}\,F = 3 \to F' = 2, 3, 4$ peaks for the cooling laser and the $^{85}\mathrm{Rb}\,F = 2 \to F' = 1, 2, 3$ peaks for the repumper (= repumping laser).

To find the MOT, we centered the cooling-laser scan on the $F = 3 \to F' = 4$ peak, shifted the offset slightly to the red, and reduced the scan amplitude so that we were scanning only a small region, red-detuned of the cooling transition. We then watched the CCD camera, which was attached to a monitor, for short flashes of a MOT as the frequency swept past the right detuning. The first attempts typically showed nothing, and we went back to improve the beam alignment, repeating the cycle until a faint MOT appeared.

Optimising and locking

Once even a faint MOT was visible, we used its brightness as the optimisation signal. We went back and forth between improving the x and y mirror alignments and retuning the cooling and repumper offsets, each time keeping whatever setting made the MOT brighter. We did not need to apply a permanent magnet to shift the zero of the magnetic field, as the as-built coil position was good.

We reduced the scan amplitudes far enough that the frequency sweep of each scan was small enough to keep the MOT visible across the entire scan, so the MOT was already stable while the laser was still being scanned over a narrow range. At this point, we first locked the cooling laser since its detuning has to be set more precisely than that of the repumper, and then we did the same for the repumper. The calibrated MOT is shown in Figure 4 and further characterized and discussed in the following section.

IV. MOT SIZE

Important note: During the analysis, it became apparent to us that we did not explicitly measure the distance d_{MOT} from the MOT to the CCD camera and the distance d_{ruler} from the mm ruler to the CCD camera. As the MOT and ruler were certainly not the same distance away from the CCD camera, the relative distances $d_{\mathrm{ruler}} = 20(10)\,\mathrm{mm}$ and $d_{\mathrm{MOT}} = 60(10)\,\mathrm{mm}$ represent mere estimates made in hindsight. As such, the uncertainties were chosen accordingly. We estimated d_{MOT} based on the distance between the MOT and photodiode used to measure the MOT fluorescence, which was roughly 50 mm. As the CCD camera was slightly further away from the vacuum chamber we added an additional 10 mm to account for this. On the other hand, the estimate of $d_{\mathrm{ruler}} = 20(10)\,\mathrm{mm}$ is made based on memory. As this is quantitatively imprecise, we reflect this as a 50% relative uncertainty. Both uncertainties are propagated to the final MOT size result. To sum up, we note, the result should be interpreted with the above in mind.

To estimate the size of the MOT, we captured images of the atomic cloud inside the vacuum chamber together with a 1 mm reference scale. The top and middle panels of Figure 4 show the raw MOT image and the ruler image, respectively. A red cross in the top panel marks the MOT, while the vertical cyan line in the middle panel illustrates the calibration measurement performed in Fiji [5] . We measured a distance of 390 pixels across a 10 mm range, yielding a ruler-plane resolution of $39.0(2)\,\frac{\mathrm{pixels}}{\mathrm{mm}}$. Because the Fiji measurement is restricted to integer pixel values, we assigned a reading uncertainty of 2 pixels over the 10 mm span, corresponding to $0.2\,\frac{\mathrm{pixels}}{\mathrm{mm}}$. This corresponded to the average deviation when performing the calibration several times by hand.

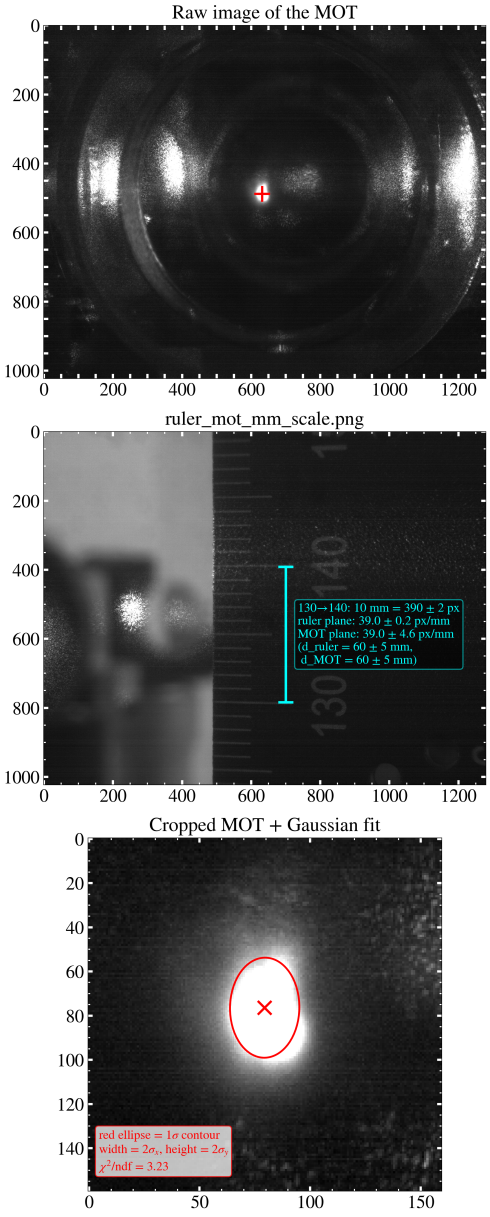


FIG. 4: Spatial characterization of the Magneto-Optical Trap (MOT). Top: Raw fluorescence image of the trapped atomic cloud inside the vacuum chamber. The red cross marks the MOT. **Middle:** Reference image of a millimeter scale used to calibrate the camera resolution at the ruler plane. The vertical cyan line indicates the measurement path, which yielded $39.0(2)\,\frac{\mathrm{pixels}}{\mathrm{mm}}$. Since the ruler image and MOT image were taken at different object distances, the calibration was transferred to the MOT plane using a distance correction factor $d_{\mathrm{ruler}}/d_{\mathrm{MOT}} = 0.333 \pm 0.176$, giving an effective MOT-plane calibration of $13.0(69)\,\frac{\mathrm{pixels}}{\mathrm{mm}}$. **Bottom:** Zoomed-in picture of the MOT fitted with a two-dimensional Gaussian function. The red ellipse denotes the 1σ contour of the fit, which was used to extract the macroscopic dimensions of the MOT.

However, the ruler image and the MOT image were not taken at the same distance from the camera. We therefore corrected the ruler-plane calibration to the MOT plane using the simple magnification scaling

$$\left(\frac{\text{px}}{\text{mm}}\right)_{\text{MOT}} = \left(\frac{\text{px}}{\text{mm}}\right)_{\text{ruler}} \frac{d_{\text{ruler}}}{d_{\text{MOT}}}$$

with $d_{\text{ruler}} = 20(10)\,\text{mm}$ and $d_{\text{MOT}} = 60(10)\,\text{mm}$. This gives a distance correction factor of

$$\frac{d_{\text{ruler}}}{d_{\text{MOT}}} = 0.33 \pm 0.18$$

and therefore an effective MOT-plane calibration of

$$\left(\frac{\text{px}}{\text{mm}}\right)_{\text{MOT}} = 13(7)\,\frac{\text{pixels}}{\text{mm}}$$

To quantitatively determine the MOT dimensions, we fitted a two-dimensional Gaussian function to the spatial profile of the cloud, as shown in the bottom panel of Figure 4 . The fit returned standard deviations of $\sigma_x = 15.68(4)\,\text{pixels}$ and $\sigma_y = 22.65(6)\,\text{pixels}$. We favor this deterministic fitting procedure over manual width estimates because it reduces subjective bias and yields robust parameter values. Visual inspection confirms that the fit captures the macroscopic horizontal and vertical widths of the MOT well, as illustrated by the red 1σ contour. The fit yields a reduced chi-squared value of $\chi^2/\text{ndf} = 3.23$ ($\chi^2 = 82727.8$, ndf $= 25593$). This value exceeds unity, indicating that the effective pixel noise is likely underestimated and that the MOT profile is not perfectly Gaussian. In particular, asymmetric deviations are visible in the lower-right and upper-right parts of the cloud, which contribute to the elevated χ^2/ndf.

Finally, we converted the fitted pixel parameters into physical dimensions and propagated the uncertainties from both the Gaussian fit and the calibration factor. In this conversion, the dominant source of uncertainty is no longer the Gaussian fit itself, but the distance-correction factor arising from the different camera-object distances. We obtained the full macroscopic width and height of the MOT by multiplying the fitted standard deviations σ_x and σ_y by a factor of 2, corresponding to the geometric diameter of the 1σ ellipse. Using the corrected MOT-plane calibration, we determined

$$\begin{aligned} \Delta x &= 2 \cdot \sigma_x = 2 \cdot 1.2(6)\,\text{mm} &&= 2.4(12)\,\text{mm} \\ \Delta y &= 2 \cdot \sigma_y = 2 \cdot 1.7(9)\,\text{mm} &&= 3.5(18)\,\text{mm} \end{aligned} \tag{1}$$

V. $\frac{\lambda}{4}$-PLATE INFLUENCE ON MOT FLUORESCENCE

To systematically investigate the influence of the quarter-wave $\left(\frac{\lambda}{4}\right)$ plate on the MOT fluorescence, we first zeroed the power meter to account for ambient background light. Because the waveplate transforms the polarization state of the trapping beams, the resulting fluorescence power $P(\theta)$ exhibits a periodic dependence on the waveplate rotation angle θ. We modeled this behavior using an adapted form of Malus's Law:

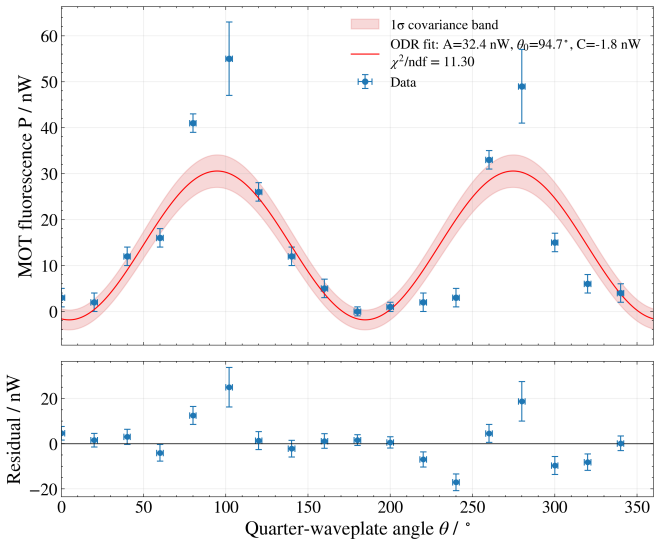


FIG. 5: Top panel: MOT fluorescence power as a function of the quarter-wave plate rotation angle θ. The solid curve represents an Orthogonal Distance Regression (ODR) fit of an adapted Malus's law to the experimental data. **Bottom panel: Fit residuals displaying the deviation of the data from the model**, highlighting systematic discrepancies near the optimum trapping angles. The red band around the fit line represents the 1σ covariance band of the fit.

$$P(\theta) = A\cos^2(\theta - \theta_0) + C \tag{2}$$

where A represents the amplitude, θ_0 accounts for the phase shift, and C provides an offset for any residual, non-zero background signal. To rigorously account for uncertainties in both the independent variable (the inherent $\pm 2°$ reading error of the rotational mount corresponding to the smallest reading increment on the $\frac{\lambda}{4}$-plate's rotational mount.) and the dependent variable (power meter fluctuations), we performed the fit using Orthogonal Distance Regression (ODR).

The top panel of Figure 5 illustrates the experimental data alongside our ODR fit, demonstrating the expected periodic behavior. The fit yielded a phase shift of $\theta_0 = 94.7(13)°$. Because this value lies close to 90°, the system effectively follows a $\sin^2(\theta)$ distribution with a small nominal residual phase shift of 4.7°. Furthermore, the fit returned a negative offset of $C = -1.8(6)\,\text{nW}$. Because negative power is unphysical, we attribute this value to a fit artifact or driven by a slow drift in the ambient background light during the measurement. This sug-

gests that dynamically re-zeroing the power meter would potentially suppress this systematic drift.

We also note that the fitted amplitude $A = 32.4(14)\,\mathrm{nW}$ fails to fully capture the experimental peak values. During the measurement, the peak fluorescence exhibited large temporal fluctuations, which is why we assigned higher uncertainties to these specific data points. Consequently, the ODR algorithm down-weighted these high-variance points, leading to a reduced weight in the fit algorithm. This can lead to a potential underestimation of the peak amplitude.

Due to the discrepancy between the fit and the peak values, the residuals of these points dominate the χ^2 sum, directly degrading the goodness-of-fit and yielding a reduced chi-squared of $\chi^2/\mathrm{ndf} = 11.30$ ($\chi^2 = 169.48$, $\mathrm{ndf} = 15$). The bottom panel of Figure 5 displays the fit residuals, visually confirming these systematic discrepancies. The data points at the fluorescence maxima, located at $102(2)^\circ$ and $280(2)^\circ$, exhibit the largest absolute deviations from the zero-line. In addition, we observe additional, seemingly-random deviations for the data points immediately surrounding these peaks. This is especially pronounced in the region $210(2)^\circ \leq \theta \leq 330(2)^\circ$. As there doesn't appear to be a clear pattern given the small amount of data, we attribute the deviations most likely to pure fluctuations in the powermeter reading. This seems to be the most likely explanation, as obtaining relatively stable data readings on the powermeter proved to be one of the main challenges during the experiment.

From the raw data, we identified the optimum operating angle for maximum MOT fluorescence to be $\theta_{\mathrm{opt.}} = 102(2)^\circ$, which we utilized for all subsequent measurements.

Finally, we investigated the retro-reflected trapping beam. We observed that rotating the $\frac{\lambda}{4}$ plates in the back-reflected beam did not alter the MOT fluorescence power beyond the standard fluctuations of the power meter. This insensitivity matches our physical expectations: in a retro-reflected geometry, the light undergoes a double-pass through the waveplate and a reflection at the mirror. This combination intrinsically reverses the propagation direction and flips the polarization handedness (e.g., from σ^+ to σ^-), ensuring the correct angular momentum configuration for the MOT regardless of angular misalignments in the retro-optics.

VI. MAGNETIC FIELD INFLUENCE ON MOT FLUORESCENCE

To investigate the influence of the magnetic field strength on the MOT fluorescence, we systematically varied the current I applied to the anti-Helmholtz coils. During the initial measurements, we observed a baseline drift in the ambient background light, which manifested as unphysical negative power readings reaching approximately $-4\,\mathrm{nW}$ when the MOT was absent. To compensate for this systematic offset, we applied a uniform $+4\,\mathrm{nW}$ correction to all recorded fluorescence values. The MOT fluorescence dependence was measured up to a maximum coil current of $I_{\mathrm{max}} = 3.01(6)\,\mathrm{A}$, as this represented the maximum achievable output current during the experiment.

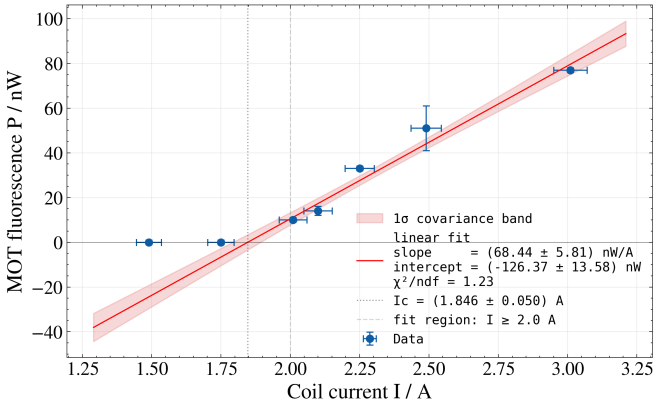


FIG. 6: MOT fluorescence power as a function of the anti-Helmholtz coil current. A global $+4\,\mathrm{nW}$ offset was applied to the data to correct for a systematic background drift. The solid line represents an Orthogonal Distance Regression (ODR) linear fit applied to the regime $I \geq 2\,\mathrm{A}$, accounting for uncertainties in both the current and the fluorescence power. The shaded region denotes the corresponding 1σ covariance band of the fit. Extrapolating this linear model to zero fluorescence yields the turn-on current $I_c = 1.85(5)\,\mathrm{A}$, representing the minimum magnetic field gradient required to initiate atom trapping.

We obtained stable current readings directly from the power supply without observable temporal fluctuations. According to the manufacturer's specifications for the device [6], the inherent measurement uncertainty for currents $I > 1\,\mathrm{A}$ is $1\% + 3$ counts. Given the device's display resolution of $10\,\mathrm{mA}$, 3 counts corresponds to an absolute uncertainty of $0.03\,\mathrm{A}$. Consequently, we used a total current uncertainty of:

$$\delta I = 0.01 \cdot I + 0.03\,\mathrm{A} \tag{3}$$

Because this intrinsic device accuracy heavily dominates the horizontal (x-axis) error budget, future iterations of this experiment would greatly benefit from utilizing a more precise ammeter. Conversely, the vertical (y-axis) uncertainties in the MOT fluorescence are barely visible for most data points because the optical power remained relatively stable during the measurement. We only observed significant transient fluctuations for the single measurement point at roughly $2.5\,\mathrm{A}$, resulting in an assigned absolute uncertainty of $10\,\mathrm{nW}$.

To accurately incorporate both the x and y uncertainties into our analysis, we utilized ODR to fit a linear function;

$$P(I) = m \cdot I + b \tag{4}$$

to the fluorescence response in the regime where $I \geq 2\,\mathrm{A}$. We expect a linear dependence in this regime because the magnetic field gradient produces a linear restoring force on the atoms; this force scales directly with the applied coil current, dictating the steady-state trapped atom number and the correspondingly emitted fluorescence.

The ODR fit returned a slope of $m = 68(6)\,\frac{\mathrm{nW}}{\mathrm{A}}$ and an intercept of $b = -126(14)\,\mathrm{nW}$. The resulting 1σ covariance band is shown as the shaded region in Figure 6, visually confirming the precision of the model within the fitted domain. From these parameters, we extrapolated the minimum coil current required to initiate atom trapping, denoted as the turn-on current I_c. By calculating the root of the linear model ($I_c = -b/m$) and propagating the errors using the full covariance matrix of the fit parameters, we determined a turn-on current of $I_c = 1.850(50)\,\mathrm{A}$.

Furthermore, the fit yielded a reduced chi-squared of $\chi^2/\mathrm{ndf} = 1.23$ ($\chi^2 = 3.70$, $\mathrm{ndf} = 3$). This corresponds to a p-value of approximately 29.7%, meaning there is a nearly 30% probability of observing this level of deviation purely by statistical chance. As a result, we conclude that the observed discrepancies between the data and the model are not statistically significant and the linear model provides a compatible description of the MOT fluorescence behavior for coil currents above 2 A.

VII. MOT LOADING BEHAVIOUR

To characterize the loading behavior of the MOT, we recorded the temporal evolution of the MOT fluorescence using a photodiode. Visual inspection of the acquired traces (see Figure 7) reveals a distinct change in the noise amplitude before and after the magnetic field is switched off (the "dump" phase). When the MOT is fully loaded, the emitted photons arrive at the photodiode independently, governed by a Poisson distribution. To account for this signal-dependent photon shot noise alongside the inherent electronic noise of our detection system, we defined a pointwise uncertainty model for the voltage variance:

$$\sigma^2(V) = A(V - V_{\mathrm{dump}}) + B \tag{5}$$

Equivalently, the statistical weight for each data point is given by $\sigma_i = \sqrt{A(V_i - V_{\mathrm{dump}}) + B}$. In this model, B explicitly represents the constant electronic and ambient background noise, corresponding to the variance exactly at the dump level ($\sigma^2(V_{\mathrm{dump}}) = B$). The coefficient A captures the signal-dependent noise contribution.

We empirically determined these parameters by analyzing distinct temporal windows within the recorded traces. From the dump regions (where the MOT is absent), we calculated a pooled dump voltage of $V_{\mathrm{dump}} = 0.0751\,\mathrm{V}$ and extracted the constant noise term $B = 0.149\,\mathrm{mV}^2$, corresponding to $\sigma_{\mathrm{dump}} = 0.387\,\mathrm{mV}$. From the fully loaded plateau regions, we computed a pooled mean voltage of $V_{\mathrm{plat}} = 0.0892\,\mathrm{V}$, a corresponding variance of $\sigma^2_{\mathrm{plat}} = 0.513\,\mathrm{mV}^2$, and a standard deviation of $\sigma_{\mathrm{plat}} = 0.716\,\mathrm{mV}$. Using these values, we derived the slope parameter:

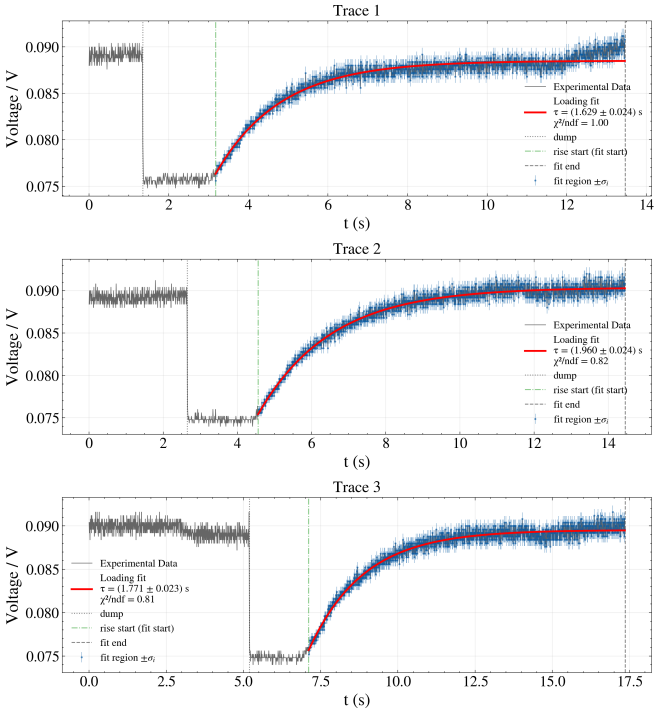


FIG. 7: Fluorescence signals recorded during the loading phase of the MOT for three traces using a photodiode. The experimental data are fitted with an exponential saturation model, further explained in Section VII , representing the physical loading of atoms in the MOT when a magnetic field is turned on (see dashed vertical green line). The visually distinct noise envelope—transitioning from the low-noise dump region to the high-noise loaded plateau—demonstrates the signal-dependent nature of the photon shot noise.

$$A = \frac{\sigma^2_{\mathrm{plat}} - B}{V_{\mathrm{plat}} - V_{\mathrm{dump}}} = 2.58 \times 10^{-5}\,\mathrm{V} \tag{6}$$

With the data correctly weighted, we applied a nonlinear least-squares fit to the MOT loading curves using an exponential saturation model:

$$V(t) = V^{\mathrm{fix}}_{\mathrm{base}} + \left(V_\infty - V^{\mathrm{fix}}_{\mathrm{base}}\right)\left(1 - e^{-(t-t_0)/\tau}\right) \tag{7}$$

We fixed the baseline voltage $V^{\mathrm{fix}}_{\mathrm{base}}$ to the mean value of the respective local dump region. Consequently, the

model relies on only $p = 3$ free parameters: the asymptotic saturation voltage of the fully loaded MOT (V_∞), the characteristic loading time constant (τ), and the effective time offset (t_0) relative to the chosen rise-start point.

To assess the goodness-of-fit, we evaluated the reduced chi-squared metric, $\chi^2_\nu = \chi^2/\mathrm{ndf}$. Because our fits encompass a large number of degrees of freedom ($\mathrm{ndf} \approx 1000$) across all traces (see Table I), the χ^2_ν distribution closely approximates a Gaussian (by the central limit theorem) with an expected mean of 1 and a standard deviation of $\sigma_\nu = \sqrt{2/\mathrm{ndf}}$. Under this framework, we can interpret deviations from unity in terms of standard confidence intervals.

For Trace 1 in Figure 7 , the obtained $\chi^2_\nu = 1.00$ matches the expected mean, seemingly validating the noise model for that specific dataset. However, the fits for Traces 2 and 3 (see Figure 7) yielded χ^2_ν values of 0.82 and 0.81, respectively. Given their specific degrees of freedom ($\mathrm{ndf} = 987$ and 1022), these correspond to distribution standard deviations of $\sigma_\nu \approx 0.045$ and 0.044. Consequently, Traces 2 and 3 deviate by 4.0σ and 4.3σ below the expected mean. Using the cumulative distribution function for the chi-squared distribution, the probabilities of observing such low χ^2_ν values purely by statistical chance (p-values) are: $p \approx 3 \times 10^{-5}$ for Trace 2 and $p \approx 8.5 \times 10^{-6}$ for Trace 3.

These statistically significant under-dispersions indicate that the data fluctuate less than predicted by our pointwise uncertainty model. A plausible explanation is temporal autocorrelation in the time-series, which would reduce the number of effectively independent samples and therefore make the nominal ndf too large. In that case, dividing the raw χ^2 by an overestimated ndf artificially suppresses χ^2_ν. Thus, while the linear variance model captures the overall shot-noise scaling, a more rigorous treatment would require an explicit autocorrelation analysis to determine the effective number of degrees of freedom. We note, however, that this is beyond the scope of this analysis.

We also observe systematic late-time deviations from the fit, most clearly in Trace 1 starting at about 12 s, and to a lesser extent in Trace 3 for times above about 15 s. To quantify this effect, we split the residual contribution to χ^2 at these times (and at 12 s also for Trace 2 for comparison), shown in Figure 13 in Appendix A . For Trace 1, the region after 12 s contains only 146 of 1029 fit points (14.2% of the data), but contributes $\chi^2 = 366.0$, i.e. 35.9% of the total $\chi^2 = 1020.9$. The mean contribution per point therefore rises from 0.742 before 12 s to 2.507 afterwards. For Trace 3, the region after 15 s contains 235 of 1025 points (22.9%) and contributes $\chi^2 = 251.3$, i.e. 30.4% of the total $\chi^2 = 825.5$, corresponding to an increase in χ^2 per point from 0.727 to 1.069. By contrast, Trace 2 does not show a comparably clear systematic upward trend in the residuals, although the late-time region after 12 s still contributes $\chi^2 = 279.7$ from 245 of 990 points (34.7% of the total). This shows that the late-time tail contributes disproportionately to the total χ^2, especially for Trace 1, and therefore highly contributes to the fit result. However, we note that a similar behaviour, of a change in noise level, is also seen in the pre-dump region of Trace 3 (see Figure 7), where the noise level trends down before stabilizing again. Therefore, the changes / trends in the noise level likely result from unknown background events.

The results of the individual loading fits are summarized in Table I. To extract a final, unified loading time, we computed the arithmetic mean of the three individual time constants, which yielded a central value of $\bar{\tau} = 1.787\,\mathrm{s}$. The uncertainty of the final value was taken as the standard error of the mean. Therefore, we determine the final characteristic MOT loading time to be

$$\tau = 1.79(10)\,\mathrm{s} \tag{8}$$

whose error is dominated by trace-to-trace scattering of the τ values rather than fit uncertainty.

VIII. RB-RB CROSS SECTION

To determine the Rb-Rb collisional cross section, denoted as σ_{Rb}, we followed the procedure outlined by Wieman et al. [7] . In this approach, we assume that the background collision rate is completely dominated by Rb-Rb interactions, treating the influence of any foreign contaminants in the vacuum chamber as negligible. This simplification allows us to directly relate σ_{Rb} to the previously determined characteristic MOT loading time, $\tau = 1.79(10)\,\mathrm{s}$ (see Section VII), via the relation: [7]

$$\sigma_{\mathrm{Rb}} = \frac{1}{n\,\bar{v}_{\mathrm{Rb}}\,\tau} \tag{9}$$

where n represents the atomic particle density and $\bar{v}_{\mathrm{Rb}}$ is the average speed of the surrounding Rb atoms in the vacuum chamber. Because the experiment operates with "low pressure of rubidium atoms" [4] , we model the Rb vapor as an ideal gas. This enables us to calculate the particle density directly from the ideal gas law:

$$n = \frac{p}{k_{\mathrm{B}} T} \tag{10}$$

where p is the chamber pressure, $k_{\mathrm{B}} = 1.380\,649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ [8] is the Boltzmann constant, and T is the absolute temperature. We recorded an ambient laboratory temperature of $T = 23.1(1)\,^\circ\mathrm{C}$ [1] , corresponding to $296.25(10)\,\mathrm{K}$.

[1] We used an estimated uncertainty of $\Delta T = 0.1^\circ$ because the thermometer had a resolution of 0.1° and the temperature drifted only slightly.

TABLE I: Loading-fit results for the three traces using the fixed-baseline exponential model shown in Equation (7) **.** A voltage-dependent uncertainty model is used which is further explained in Section VII .

Panel	$V_{\mathrm{base}}^{\mathrm{fix}}$/V	V_∞/V	τ/s	t_0/s	ndf	χ^2_ν
Top	$7.566(5)\times10^{-2}$	$8.851(3)\times10^{-2}$	1.629(24)	−0.089(11)	1026	1.00
Middle	$7.486(5)\times10^{-2}$	$9.037(4)\times10^{-2}$	1.960(24)	−0.074(10)	987	0.82
Bottom	$7.487(5)\times10^{-2}$	$8.953(4)\times10^{-2}$	1.771(23)	−0.098(10)	1022	0.81
Combined			1.78(10)			

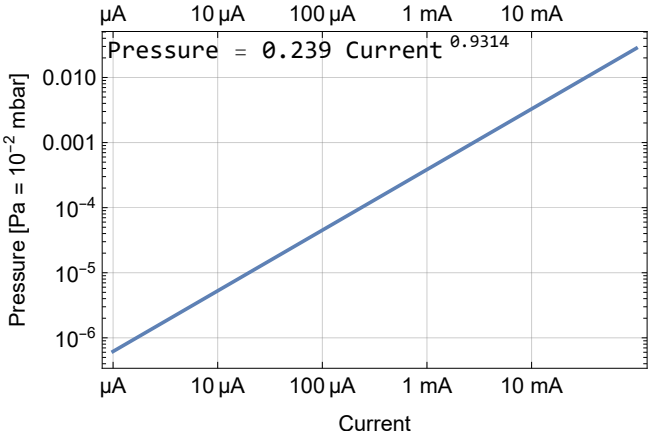


FIG. 8: Calibration curve for the ion pump, detailing the conversion from pump current to vacuum chamber pressure. The function $p(I) = 0.239 \cdot I_{\mathrm{pump}}^{0.9314}$ yields the pressure in Pascals when providing the current in Amperes. This calibration data was provided by the MOT laboratory instructors [9].

To determine the pressure p, we read the current from the ion pump controller, obtaining $I_{\mathrm{pump}} = 1.00 \times 10^{-2}$ mA. According to the manufacturer's specifications [10] , the nominal reading uncertainty is 0.02%. Because this reading error leads to an error that is four orders of magnitude smaller than the measurement value itself, we treat the recorded current as effectively exact. Furthermore, we omit any temperature-related measurement uncertainties for the pump electronics; the manufacturer specifies an error of 0.05% per Kelvin [10] , which likely yields deviations in the per-mille range (relative to the measurement value), which is why we neglect it. Using the calibration function provided in Figure 8 , we converted the current ($I_{\mathrm{pump}} = 1.00\times10^{-5}$ A) to an absolute pressure of $p \approx 5.265 \times 10^{-6}$ Pa. Inserting this pressure and our measured temperature into the ideal gas law yields a particle density of $n = 1.287(1) \times 10^{15}\,\mathrm{m}^{-3}$, where the minor uncertainty propagates strictly from the room temperature reading.

Because the vacuum chamber approximately constitutes a closed system operating as an ideal gas, the thermal speed motion of the atoms follows a Maxwell-Boltzmann distribution. To calculate the effective probability of Rb-Rb interactions, we use the mean speed of the ensemble, which represents the average path length traveled per unit time, rather than the most probable speed. [2]

The mean speed $\bar{v}_{\mathrm{Rb}}$ is given by: [11]

$$\bar{v}_{\mathrm{Rb}} = \sqrt{\frac{8k_{\mathrm{B}}T}{\pi\, m_{\mathrm{Rb}}}} \tag{11}$$

where m_{Rb} is the mass of the ^{85}Rb isotope. To convert the given atomic mass ($m_{\mathrm{Rb}} = 84.911\,789\,737\,9(54)$ u [12]) to kilograms, we multiply it by the atomic mass constant $m_u = 1.660\,539\,066\,60(50) \times 10^{-27}$ kg [13] . To account for the uncertainty during this conversion, we added the relative uncertainties of both constants in quadrature, which we then multiplied by the nominal mass in kilograms. This calculation yielded $m_{\mathrm{Rb}} = 1.409\,993\,44(43) \times 10^{-25}$ kg. Inserting this mass and the ambient temperature into the speed distribution gives an average atomic speed of $\bar{v}_{\mathrm{Rb}} = 271.79(5)$ m/s.

Finally, we synthesized these parameters to evaluate the collisional cross section. By inserting the calculated atomic density n, the mean speed $\bar{v}_{\mathrm{Rb}}$, and the loading time constant $\tau = 1.79(10)$ s, we determined the Rb-Rb cross section to be:

$$\sigma_{\mathrm{Rb}} \approx 1.60(9) \times 10^{-18}\,\mathrm{m}^2 = 1.60(9) \times 10^{10}\,\mathrm{b} \tag{12}$$

We note that the final uncertainty of $0.09\times10^{-18}\,\mathrm{m}^2$ is dominated by the 5.6% relative error in the MOT loading time τ, which overshadows the uncertainties contributed by the temperature measurement and fundamental constants.

We note that, as the concrete Rb-Rb cross section depends on the obtained MOT and is not a fundamental constant, a direct comparison to other literature results is hardly possible.

[2] We utilize the mean speed, rather than the most probable speed, because the collision probability directly depends on the average distance the background atoms travel per unit time (the mean free path length), which is governed by the integrated average of the Maxwell-Boltzmann distribution.

IX. LASER FREQUENCY DETUNING

We measured the MOT fluorescence and laser spectrum during a slow frequency scan of the cooling laser while the repumping laser remained locked. This was achieved by feeding a triangle signal into the laser piezo. The goal was to determine the optimal detuning of the cooling laser relative to the ^{85}Rb $F = 3 \rightarrow F' = 4$ cooling transition.

A. Frequency Calibration

To convert the time axis to a frequency axis, we fitted Lorentzian profiles to three peaks in the spectrum shown in Figure 9. These peaks were:

1. CO(2,4) crossover resonance at $t = 39.64(6)\,\mathrm{s}$, corresponding to $-92.02\,\mathrm{MHz}$
2. CO(3,4) crossover resonance at $t = 46.12(3)\,\mathrm{s}$, corresponding to $-60.32\,\mathrm{MHz}$
3. F'=4 hyperfine transition at $t = 58.85(18)\,\mathrm{s}$, corresponding to $0\,\mathrm{MHz}$ (the reference frequency)

To evaluate the quality of the calibration fits, we examined the reduced chi-squared metric, $\chi^2_\nu = \chi^2/\mathrm{ndf}$. For the CO(2,4) crossover, the fit yields $\chi^2_\nu = 1.1$, indicating statistical agreement between the experimental data and the standard Lorentzian model. However, the goodness-of-fit degrades for the remaining peaks, rising to $\chi^2_\nu = 2.5$ for the CO(3,4) peak and $\chi^2_\nu = 4.1$ for the $F' = 4$ transition.

Visual inspection of the spectrum in Figure 9 reveals the physical origin of this discrepancy. While the CO(2,4) peak is relatively well-isolated, the $F' = 4$ transition (and to a lesser extent, the CO(3,4) peak) exhibits a greater line broadening. This skewed lineshape is potentially not fully captured by a purely symmetric Lorentzian profile. For instance, to better capture the $F' = 4$, we additionally used a linear underground model, to increase the fit accuracy.

The three peaks of the spectrum in Figure 9 give us a linear relationship between scan time and laser frequency. The corresponding linear fit depicted in Figure 10 yielded:

$$\nu(t) = (4.886(3)\,\mathrm{MHz\,s^{-1}})\,t - 285.7(1)\,\mathrm{MHz} \qquad (13)$$

where the error bars in both x and y effectively invisible due to their small size. The residuals of the fit are $-0.03\,\mathrm{MHz}$, $0.00\,\mathrm{MHz}$, and $-1.87\,\mathrm{MHz}$; together with 3 degrees of freedom (ndf) the goodness-of-fit is $\chi^2/\mathrm{ndf} = 3.49/3 \approx 1.16$. The resulting p-value for such a deviation from $\chi^2/\mathrm{ndf} = 1$ is $p \approx 32.21\%$. Therefore, we conclude the linear model is compatible with the data, confirming that the piezo response was linear over the scan range.

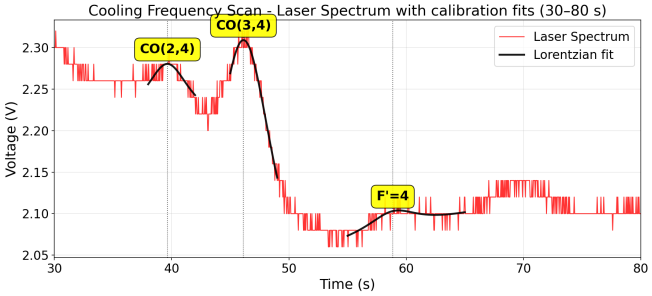


FIG. 9: Laser spectrum (CH2) during the cooling-laser frequency scan., The three Lorentzian fits to the CO(2,4), CO(3,4), and $F' = 4$ calibration features over the 35–65 s window are shown. For CO(2,4) we get $\chi^2/\mathrm{ndf} = 1.1$, for CO(3,4) $\chi^2/\mathrm{ndf} = 2.5$ and for $F' = 4$ $\chi^2/\mathrm{ndf} = 4.1$

B. MOT fluorescence peak

The MOT fluorescence signal was fitted with a Lorentzian function to determine the peak position and width 11. The fit yielded:

$$\begin{aligned} t_{\mathrm{peak}} &= 56.73(2)\,\mathrm{s} \\ \mathrm{FWHM} &= 0.80(7)\,\mathrm{s} \end{aligned} \qquad (14)$$

The Lorentzian-profile fit yields a goodness-of-fit of $\chi^2/\mathrm{ndf} = 3.8$. This is likely explained by the deviations of the data from the ideal Lorentzian-profile, specifically for the time region around 54 s, where the data is consistently above the curve, and 58 s, where the data is consistently below the curve. However, by visual inspection the peak value is contained within by fit. All other uncertainties of the fit is reflected in the fit parameters in Equation (14) .

Using the calibration function shown in (13) , the peak time corresponds to an optimal cooling-laser detuning of:

$$\delta_{\mathrm{opt}} = -8.5(1)\,\mathrm{MHz} \qquad (15)$$

relative to the $F = 3 \rightarrow F' = 4$ transition.

C. Comparison with theory

The natural linewidth of the $5P_{3/2}$ excited state of ^{85}Rb is $\Gamma = 1/(2\pi\tau) \approx 6.07\,\mathrm{MHz}$, where $\tau = 26.24\,\mathrm{ns}$ is the lifetime [14] . Our measured detuning corresponds to approximately $\delta_{\mathrm{opt}}/\Gamma \approx -1.40(1)$ natural linewidths. This is compatible with the script recommending locking the laser at "about a linewidth to the red of the transition" [4] .

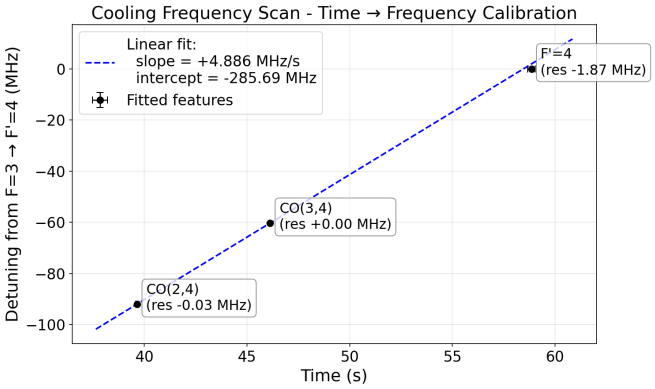


FIG. 10: Linear time-to-frequency calibration. The three fitted peak times are plotted against their known hyperfine frequencies, with error bars from the Lorentzian fits. The dashed line is the best-fit line used to convert the time axis to detuning. The error bars in both x and y values are not visible due to their small size. $\chi^2/\mathrm{ndf} = 1.16$

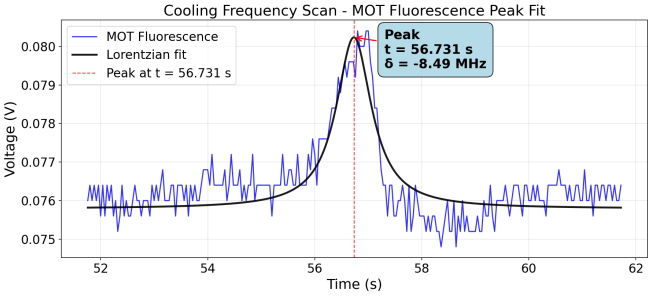


FIG. 11: MOT fluorescence peak (CH1) fitted with a Lorentzian function. The peak occurs at $t_{\mathrm{peak}} = 56.73(2)\,\mathrm{s}$, corresponding to an optimal cooling detuning of $\delta_{\mathrm{opt}} = -8.5(1)$ MHz. $\chi^2/\mathrm{ndf} = 3.8$

X. MOT POPULATION

To determine the number of atoms trapped in the MOT, we measured the fluorescence emitted by the trapped atoms and used the relationship between total emitted light and the per-atom photon scattering rate.

A. Measurement Setup

We measured the MOT fluorescence using a power-meter placed at distance $L = 49(5)\,\mathrm{mm}$ from the trap, imaging the MOT through a lens of diameter $25.4(3)\,\mathrm{mm}$. The measured fluorescence power was $P_{\mathrm{MOT}} = 50(6)\,\mathrm{nW}$ (background-subtracted by switching the magnetic field on and off). The cooling laser was detuned by $\Delta = -8.5(1)\,\mathrm{MHz}$ from the $^{85}\mathrm{Rb}$ $F = 3 \rightarrow F' = 4$ cooling transition.

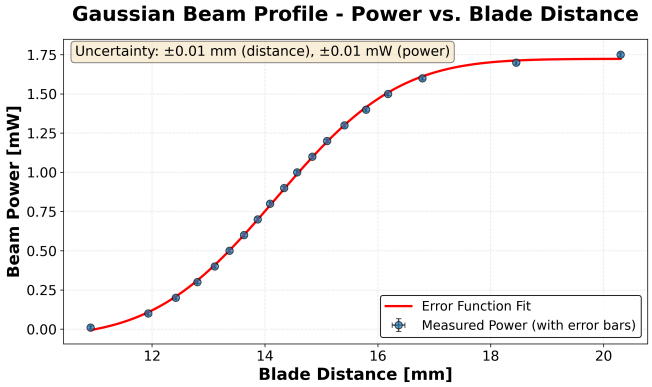


FIG. 12: Cooling laser beam profile measured using a razor-blade scan. The beam power is plotted as a function of blade position. The blue points are measurements; the red curve is the error function fit. The error bars are invisible due to their relatively small size. The fit yields $\chi^2/\mathrm{ndf} = 1.45 \times 10^{-4}$.

B. Beam Characterization

The cooling laser beam diameter was measured using a razor blade mounted on an optical rail, as described by Wieman *et al.* [7], pp. 321–322. We recorded the beam power as a function of blade position and fitted an error function the data. Figure 12 shows the measured beam power versus blade position. We note that the error bars in both x and y are virtually invisible in the fit due to their size.

The data were fitted with an error function of the form:

$$P(x) = A\,\mathrm{erf}(B(x - C)) + D \tag{16}$$

with fitted parameters:

$$A = 0.888(9)\,\mathrm{mW} \tag{17}$$

$$B = 0.418(8)\,\mathrm{mm}^{-1} \tag{18}$$

$$C = 14.20(2)\,\mathrm{mm} \tag{19}$$

$$D = 0.836(7)\,\mathrm{mW} \tag{20}$$

The resulting $\chi^2/\mathrm{ndf} = 1.45 \times 10^{-4}$ which indicates that either the errors are severely underestimated or the model fits the data extremely closely. Since we measured with a nonius, we were able to achieve the given precision of up to 0.01 mm. Therefore the likely cause for the small χ^2/ndf is the close tracking of the fit function to the data points, which renders the model-to-data deviations to virtually 0, thereby decreasing the χ^2/ndf. This is evident by the inspected residuals.

For a Gaussian beam, the error function fit parameter B is related to the $1/\mathrm{e}^2$ beam radius by $B = \sqrt{2}/w$. Therefore:

$$w = \frac{\sqrt{2}}{B} = \frac{\sqrt{2}}{0.418(8)\,\mathrm{mm}^{-1}} = 3.38(20)\,\mathrm{mm} \tag{21}$$

The corresponding beam area ($1/e^2$ definition) is:

$$A_b = \pi w^2 = \pi(3.38(20)\,\mathrm{mm})^2 = 0.360(43)\,\mathrm{cm}^2 \quad (22)$$

This beam waist is used in subsequent calculations of the laser intensity at the MOT.

C. Laser Intensity and Saturation Parameter

The six MOT beams (three orthogonal pairs) overlapped at the trap center. We measured the power in each incoming beam: $P_x = 1.127(3)\,\mathrm{mW}$, $P_y = 1.050(3)\,\mathrm{mW}$, and $P_z = 2.600(50)\,\mathrm{mW}$. The cooling laser comprises a fraction $\eta = P_{\mathrm{cool}}/P_{\mathrm{tot}} = 6.94/9.60 = 0.723(1)$ of the total light. The cooling power per axis was therefore $P_x^c = 0.815(2)\,\mathrm{mW}$, $P_y^c = 0.759(2)\,\mathrm{mW}$, and $P_z^c = 1.880(36)\,\mathrm{mW}$. For a Gaussian beam with power P and $1/e^2$ radius w, the peak intensity (at the beam center) is:

$$I_{\mathrm{peak}} = \frac{2P}{\pi w^2} \quad (23)$$

This gives peak intensities of $I_x = 4.53(54)\,\mathrm{mW/cm^2}$, $I_y = 4.22(50)\,\mathrm{mW/cm^2}$, and $I_z = 10.46(125)\,\mathrm{mW/cm^2}$ per beam. The total intensity from all six beams (two per orthogonal direction, accounting for retroreflection) is:

$$I_{\mathrm{tot}} = 2(I_x + I_y + I_z) = 38.4(46)\,\mathrm{mW/cm^2} \quad (24)$$

The saturation parameter is defined as $s_0 = I/I_s$, where $I_s = 4.1\,\mathrm{mW/cm^2}$:

$$s_0 = \frac{I_{\mathrm{tot}}}{I_s} = 9.37(112) \quad (25)$$

D. Per-Atom Photon Scattering Rate

The rate at which a single atom scatters photons depends on the light intensity, saturation, and detuning. Wieman *et al.* provide the formula:

$$R = \frac{(I/I_s)\pi\Gamma}{1 + I/I_s + 4(\Delta/\Gamma)^2} \quad (26)$$

With $\Delta/\Gamma = 8.5(1)\,\mathrm{MHz}/6.07\,\mathrm{MHz} = 1.400(16)$, we have $4(\Delta/\Gamma)^2 = 7.84(18)$. The denominator becomes $1 + (9.37(112)) + (7.84(18)) = 18.21(113)$. Therefore:

$$\begin{aligned} R &= \frac{(9.37(112)) \cdot \pi \cdot 6.07 \times 10^6\,\mathrm{Hz}}{18.21(113)} \\ &= 9.81(37) \times 10^6\,\mathrm{s^{-1}}\ \text{(photons/s per atom)} \end{aligned} \quad (27)$$

E. Photon Collection and Solid Angle

The fluorescence collected by the lens covers a solid angle Ω at the MOT. For a circular lens of radius r at distance L from the trap, the exact solid angle is:

$$\Omega = 2\pi\left[1 - \frac{L}{\sqrt{L^2 + r^2}}\right] \quad (28)$$

With $L = 49(5)\,\mathrm{mm}$ and $r = 12.70(13)\,\mathrm{mm}$, this gives $\Omega = 0.201(39)\,\mathrm{sr}$. Assuming isotropic emission from the atoms, the fraction of total emitted light collected is:

$$f = \frac{\Omega}{4\pi} = 1.60(31) \times 10^{-2} \quad (29)$$

The photon energy at 780 nm is $E_\gamma = hc/\lambda = 2.55 \times 10^{-19}\,\mathrm{J}$. The number of photons per second detected by the powermeter is:

$$\dot{n}_{\mathrm{pm}} = \frac{P_{\mathrm{MOT}}}{E_\gamma} = \frac{50(6) \times 10^{-9}\,\mathrm{W}}{2.55 \times 10^{-19}\,\mathrm{J}} = 1.96(24) \times 10^{11}\,\mathrm{s^{-1}} \quad (30)$$

The total number of photons emitted by all atoms per second is:

$$\dot{n}_{\mathrm{tot}} = \frac{\dot{n}_{\mathrm{pm}}}{f} = \frac{1.96(24) \times 10^{11}\,\mathrm{s^{-1}}}{1.60(31) \times 10^{-2}} = 1.23(28) \times 10^{13}\,\mathrm{s^{-1}} \quad (31)$$

F. MOT Atom Number

The number of trapped atoms is found by dividing the total photon emission rate by the per-atom scattering rate:

$$\begin{aligned} N &= \frac{\dot{n}_{\mathrm{tot}}}{R} \\ &= \frac{1.23(28) \times 10^{13}\,\mathrm{s^{-1}}}{9.81(37) \times 10^6\,\mathrm{s^{-1}}} \\ &\approx 1.3(3) \times 10^6\,\mathrm{atoms} \end{aligned} \quad (32)$$

G. Uncertainty Analysis

The contributions to the final uncertainties found using gaussian error propagation are summarized in Table II.

We note that there is no universal "standard" atom number for a MOT, as the total trapped population depends heavily on specific experimental parameters such as laser power, beam diameter, background Rb density, and magnetic field gradients. Therefore, a direct comparison to a universal literature value is impossible.

TABLE II: Contributions to the uncertainty in the MOT atom number. The collection solid angle dominates.

Source	Relative uncertainty	Share of $(\sigma_N/N)^2$
P_{MOT}	12.0 %	27 %
f (via L)	19.4 %	70 %
R (via w, P_{cool}, Δ)	3.8 %	3 %
σ_N/N	23.1 %	

To evaluate the reliability of this measurement, we turn to the uncertainty budget presented in Table II . The total relative uncertainty of the atom number is $\approx 23.1\%$. An analysis of the individual contributions via Gaussian error propagation reveals that this final error is dominated by the solid angle calculation (f), which alone accounts for 70% of the total variance in the measurement.

The physical origin of this contribution is the distance L between the imaging lens and the trapped atomic cloud. In our setup, this distance ($L = 49(5)\,\mathrm{mm}$) is difficult to measure accurately because the MOT is suspended in vacuum inside a glass cell, preventing direct mechanical contact with a ruler. The required visual estimation through the glass walls introduces a systematic uncertainty ($\approx 10\%$ relative error on L). Because the collection solid angle scales inversely with the square of the distance ($\Omega \propto L^{-2}$ in the far-field limit), this 10% linear error propagates quadratically, resulting in a nearly 20% uncertainty in the detected photon fraction f.

The second largest contributor is the reading error on the photodiode power measurement (P_{MOT}), accounting for 27% of the variance. This error arises primarily from ambient light fluctuations and intrinsic electronic noise in the low-power (nW) regime, which complicate the background subtraction process.

In contrast, the internal parameters characterizing the laser, such as the beam waist w, optical power P_{cool}, and detuning Δ, contribute a mere 3% to the total variance. The per-atom scattering rate R is robust against these uncertainties because the MOT operates deep within the saturated regime ($s_0 \approx 9.4$). When $I \gg I_s$, the scattering rate asymptotically approaches the maximum physical limit defined by the natural linewidth ($\Gamma/2$). Consequently, fluctuations in the laser intensity have an almost negligible effect on the rate at which an individual atom scatters photons.

Therefore, any future efforts to improve the precision of the MOT population measurement should focus on the detection geometry.

XI. CONCLUSION

We set up and characterized a magneto-optical trap for ^{85}Rb atoms. After aligning the two horizontal MOT beams and locking the cooling and repumping lasers via polarization spectroscopy, we obtained a stable MOT that served as the basis for all further measurements.

From fluorescence images we determined the cloud size to $\Delta x = 2.4(12)\,\mathrm{mm}$ and $\Delta y = 3.5(18)\,\mathrm{mm}$ (see Section IV). The uncertainties are dominated by the distance correction factor needed to account for the different camera-object distances when calibrating the ruler and MOT images. The cloud is visibly elongated along the vertical axis.

The dependence of the fluorescence on the $\lambda/4$ plate angle followed an adapted Malus's law with an optimum angle of $\theta_{\mathrm{opt}} = 102(2)^\circ$ (see Section V). The reduced chi-squared of $\chi^2_\nu = 11.30$ is poor, mainly because the peak fluorescence values fluctuated strongly during the measurement and were down-weighted by the ODR fit. As expected, rotating the waveplates in the retro-reflected beam had no measurable effect, which is consistent with the double-pass argument that automatically restores the correct circular polarization.

For the magnetic field dependence we found a clearly linear behaviour for $I \geq 2\,\mathrm{A}$ and extracted a turn-on current of $I_c = 1.85(5)\,\mathrm{A}$ (see Section VI). With $\chi^2_\nu = 1.23$ ($p \approx 29.7\%$) the linear model is statistically compatible with the data in the fitted regime.

From three loading curves we obtained a characteristic loading time of $\tau = 1.79(10)\,\mathrm{s}$, where the uncertainty is dominated by the trace-to-trace scatter rather than by the individual fit errors (see Section VII). Two of the three traces gave χ^2_ν values significantly below unity, which we attribute to temporal autocorrelation in the photodiode signal that effectively inflates the nominal number of degrees of freedom. A proper autocorrelation analysis was beyond the scope of this report. Combining τ with the chamber pressure ($p \approx 5.27 \times 10^{-6}\,\mathrm{Pa}$) and the room temperature, we obtained a Rb-Rb collisional cross section of $\sigma_{\mathrm{Rb}} = 1.60(9) \times 10^{-18}\,\mathrm{m}^2$, whose relative uncertainty is essentially inherited from τ (see Section VIII).

Scanning the cooling laser frequency, the MOT fluorescence peaked at $\delta_{\mathrm{opt}} = -8.5(1)\,\mathrm{MHz}$, corresponding to approximately $1.4\,\Gamma$ red of the $F = 3 \to F' = 4$ transition. This is compatible with the script's recommendation to lock the laser at about a linewidth to the red of the transition (see Section IX).

From the fluorescence power, the beam profile and the collection geometry we estimated the atom number to $N = (1.3 \pm 0.3) \times 10^6$ (see Section X). About 70% of the variance comes from the distance between the collection lens and the trap (which enters through the solid angle), and approximately 27% from the measured fluorescence power itself. This indicates that improving the accuracy of geometric measurements is the main way to improve the result's accuracy.

XII. ACKNOWLEDGMENTS

We acknowledge the use of Gemini 3.1 [15] for drafting support, text refinement, and LaTeX formatting assistance during the preparation of this manuscript, as well as coding support from Claude Code from Anthropic [16] and Codex [17] from OpenAI.

Appendix A: Residuals of loading behaviour fits

Figure 13 shows the residuals of the three loading-curve fits as a function of time, discussed in Section VII. The late-time points are highlighted separately in orange to emphasize the systematic upward trend visible in Trace 1 and, more weakly, in Trace 3, while Trace 2 remains comparatively structureless.

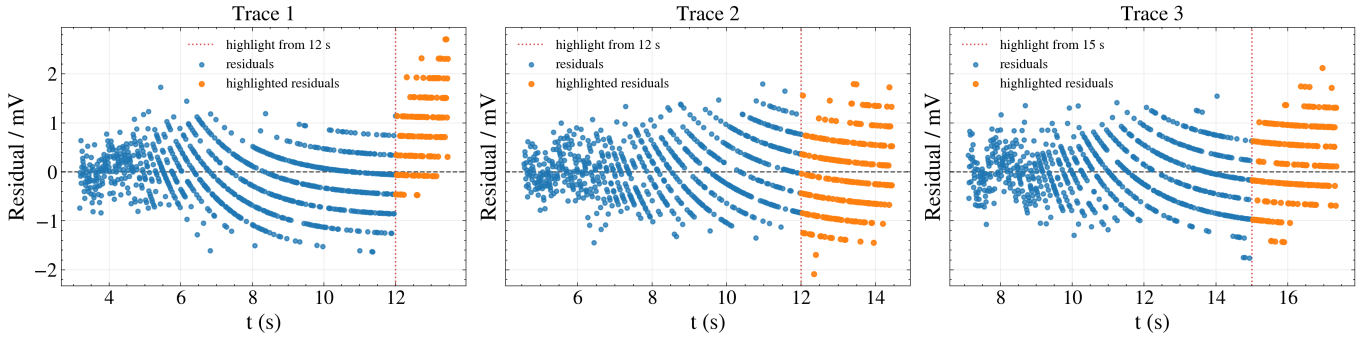


FIG. 13: Residuals of the loading-curve fits for the three traces, shown without error bars for visual clarity. Points in the late-time regions ($t > 12\,\mathrm{s}$ for Traces 1 and 2, and $t > 15\,\mathrm{s}$ for Trace 3) are highlighted in orange. The uncertainties are omitted here to reduce visual clustering and make systematic trends in the residuals easier to identify. Trace 1 and Trace 3 show a clear upward trend for highlighted region in orange, whereas Trace 2 stays comparatively flat.

[1] D. A. Steck, Rubidium 85 d line data, Available online at `http://steck.us/alkalidata` (revision 2.2.3, 9 July 2021), accessed: 2026.

[2] H. J. Metcalf and P. van der Straten, *Laser Cooling and Trapping*, 1st ed. (Springer-Verlag, New York, 1999).

[3] E. L. Raab, M. Prentiss, A. Cable, S. Chu, and D. E. Pritchard, Trapping of neutral sodium atoms with radiation pressure, Physical Review Letters **59**, 2631 (1987).

[4] Institut für Angewandte Physik (IAP), *FP Experiment: Rubidium MOT Tutorial*, University of Bonn (2014), aLC-Instructions MOT 2025 (A248 Material).

[5] Fiji Contributors, Fiji downloads, `https://imagej.net/software/fiji/downloads` (2026), accessed: 2026-04-23.

[6] Voltcraft, *Voltcraft LRP-1363 Operating Instructions Manual*, Voltcraft, Hirschau, Germany (2012), accessed: November 10, 2026.

[7] C. E. Wieman, G. E. Flowers, and S. L. Gilbert, Inexpensive laser cooling and trapping experiment for undergraduate laboratories, American Journal of Physics **63**, 317 (1995).

[8] National Institute of Standards and Technology, Codata recommended values of the fundamental physical constants: Boltzmann constant (2018).

[9] MOT Lab Instructors, Ion pump calibration data and additional mot resources (2025).

[10] Heinzinger electronic GmbH, *LNC High Voltage Power Supply Datasheet*, Heinzinger electronic GmbH, accessed: 2026-04-26.

[11] LibreTexts, 27.3: The distribution of molecular speeds is given by the maxwell-boltzmann distribution, `https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Physical_Chemistry_(LibreTexts)/27%3A_The_Kinetic_Theory_of_Gases/27.03%3A_The_Distribution_of_Molecular_Speeds_is_Given_by_the_Maxwell-Boltzmann_Distribution`, accessed: 2026-04-26.

[12] National Institute of Standards and Technology, Atomic weights and isotopic compositions for rubidium (2015).

[13] National Institute of Standards and Technology, Codata recommended values of the fundamental physical constants: atomic mass constant (2018).

[14] D. A. Steck, Rubidium 85 d line data (2023), version 2.2.1, available at http://steck.us/alkalidata.

[15] Gemini Team, Google, *Gemini 3.1 Pro: Technical Report and Model Specifications*, Tech. Rep. (Google DeepMind, 2026) technical report detailing reasoning benchmarks and agentic workflows for the Gemini 3.1 series.

[16] Anthropic PBC, Anthropic — ai research and products that put safety at the frontier (2026), accessed: 2026-04-16.

[17] OpenAI, Codex — ki-coding-partner von openai (2026), accessed: 2026-04-16.